

## Key Outcomes

The team aligned on **three core features** for the student event discovery app: last-minute pricing, transportation routing with cost calculation, and an organizer dashboard.

Technical decisions finalized **FastAPI** for backend, **OpenStreetMap** for routing, and a **heat map approach** for neighborhood busyness prediction using historical NYC datasets (subway, taxi, bike-sharing).

## Decisions Made

- **Backend framework:** FastAPI with Postgres database and Railway for deployment
- **Routing:** OpenStreetMap API instead of Google Maps, supporting multiple transport modes (subway, bike-sharing, walking)
- **Busyness prediction:** Historical data model using NYC subway ridership, taxi trips, and bike-sharing data to predict neighborhood crowdness with heat map visualization
- **Event data sources:** NYC permitted events dataset plus Ticketmaster/Eventbrite API calls for live listings
- **Pricing model:** Event organizers set last-minute prices directly; no price comparison needed as assumption is prices already discounted
- **Cost calculation:** Simple additive model (event price + transport costs) using current transit fares and bike-sharing rates
- **Rating system:** Implement user ratings to address quality concerns instead of price comparison

## Technical Architecture

### Data Sources Confirmed:

- NYC permitted events (historical and upcoming)
- NYC subway ridership (150M+ entries)
- NYC taxi trip records
- Citi Bike sharing data (rides and availability)
- Noise complaints (supplementary correlation data)

### Model Approach:

- Regression or decision tree model predicting low/medium/high busyness ranges for neighborhoods
- Heat map overlay showing probability of finding events by area and time
- Historical patterns (e.g., Friday 8PM = 500K subway riders vs 50K baseline = high busyness)

### API Integration:

- Ticketmaster and Eventbrite for event listings (rate limits acknowledged)
- OpenStreetMap Overpass API for venue database and routing
- Weather API consideration for busyness correlation

## Feature Specifications

### Mobile vs Web Differentiation:

- **Mobile:** Primary user interface for students discovering events
- **Web:** Organizer dashboard for listing management
- Xiya (Product/UX Lead) to define detailed feature breakdown

### User Stories Supported:

- Students finding last-minute events within 1-2 hours
- Cost-conscious routing with bike/subway options for budget optimization
- Discovery mode: "Where should I go to find spontaneous activities?"
- Organizers posting last-minute tickets to fill capacity

## Pending Confirmation

- **Busyness visualization:** Final decision between heat map, badges, or numerical predictions (leaning toward heat map)
- **Database schema:** Dan to design and share with team for frontend/backend coordination
- **DEI integration:** Accessibility features in routing to be specified
- **Time-series modeling:** Whether to continuously update models or run once with static training data
- **API rate limits:** Ticketmaster restricted to 10-20 calls/day; need alternative or caching strategy
- **Too Good To Go integration:** API availability unclear; Dan to investigate

## Research & User Validation

### Interview Plan:

- Xiya to create questionnaire for distribution to student networks
- Each team member to identify 2-3 students for iterative interviews across development stages (pre-mockup, post-mockup, prototype, final)
- Data scraping from Reddit and online student communities for leisure preferences
- Multiple interview rounds: before mockups, after mockups, with prototype, post-prototype

## Team Coordination

- **Next meeting:** Tuesday (full team sync)
- **Communication:** Clear documentation required as parallel development begins on frontend, backend, and data modeling
- **Team contract:** Submitted to Ferrari

## Todo Items

- **Dan:** Set up FastAPI backend structure, design database schema, share framework documentation with team (by Tuesday)
- **Erwan:** Begin exploratory analysis on NYC subway, taxi, and bike datasets; validate data quality and merge feasibility
- **Xiya:** Create and distribute student questionnaire; define mobile vs web feature differentiation; prepare UX mockups
- **All team members:** Track time in budgetplan sheet including research and meeting prep